

Anisotropic sources for surface and volume boundary layer mesh generation

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ABSTRACT

A new approach to model and compute the metric tensor corresponding to *a-priori* prescribed surface and volume boundary layers is presented. This is achieved by the use of sources that model the combined anisotropic metric. Algorithms are presented to both evaluate the resulting metric value at any point in space and the efficient approximation of the minimum and maximum metric over an axis-aligned box which enables fast evaluation of the anisotropic metric independent of the approach used for mesh generation. The ability to model and evaluate the anisotropic sizing independently ensure that mesh generation algorithms that use it can produce meshes that correctly and predictably capture the sizing fields induced by the user prescribed boundary layer attributes. The method is unique in producing globally smooth representations of the final anisotropic metric including the influence of implicit anisotropy due to boundary curvature and any prescribed isotropic sizes. As a result, it addresses robustness and size smoothness issues in legacy approaches that decouple the surface and volume mesh generation stages. The value of the new method is demonstrated by several examples that include both surface and volume boundary layers prescribed on geometries with curved boundaries.

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1. Introduction

In order to numerically resolve physical phenomena which present strong spatial anisotropic features such as boundary layers, shear-layers, or shock waves, meshes with strong spatial anisotropy are required. For anisotropic solution features that are not known in advance, *a-posteriori* adaptation of a starting mesh is often used to produce anisotropic meshes that resolve such features. For flow-physics with boundary layers, one can generate anisotropic meshes to capture the boundary layer *a-priori* with strong size gradients normal to the boundaries. Following early pioneering work [18,24,29], *a-priori* boundary layer mesh generation has become a research topic of its own due to the technical challenges posed by complex geometries requiring the resolution of multiple boundary layers interacting with tight geometric constraints [2,22].

Robust¹ generation of good quality² boundary layer meshes for complex geometries requires addressing the following key technical issues:

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¹ Refers to successful generation of a mesh.

² Entails, at a minimum, satisfying the anisotropy, resolving boundary geometry and ensuring strictly positive element volumes.

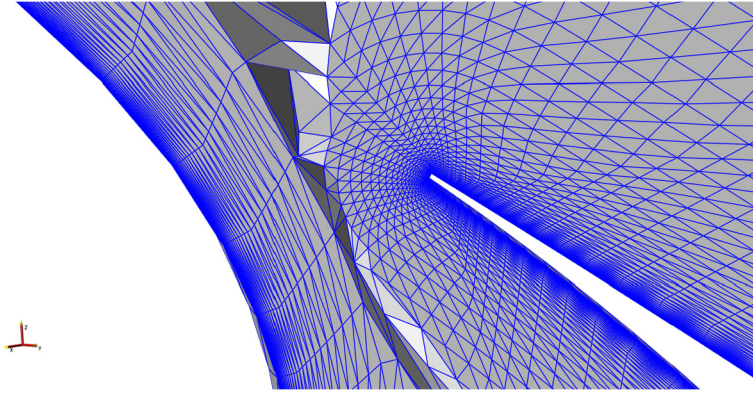


Fig. 1.1. Close-up view showing the interaction between boundary layers growing from the trailing edge of the slat and the leading edge of the wing in a multi-element airfoil. Due to proximity, the outer faces of the wing have not reached isotropy, and a jump in size appears. This is due to the fact that the anisotropy of the boundary layer of one element has not been taken into account when surface meshing the other.

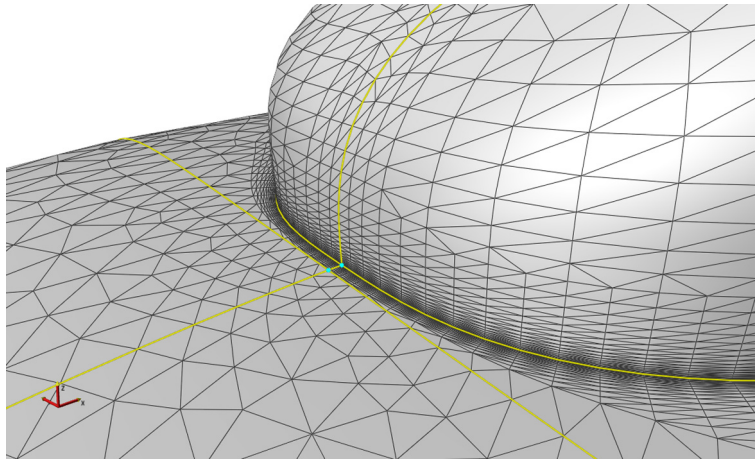


Fig. 1.2. The surface boundary layer has no space to fully reach isotropy because of a geometric edge. The other face does not receive any anisotropic attribute and grows isotropically, generating a size jump.

1. Resolve interactions of multiple boundary layers due to complex boundary geometry in close vicinity.
2. Maintain smooth variation of the mesh size spatially in all directions.

The last issue is critical for solution accuracy and convergence as discretization (interpolation) schemes require a measure of smoothness in size variation for the underlying mesh. Figs. 1.1 and 1.2 illustrate situations that lead to non-smooth size variation due to improper handling of interacting boundary layers and not accounting for constraints posed by boundary topological features, respectively.

Most existing approaches to volume boundary meshing start with the generation of an isotropic surface mesh of the boundary followed by the generation of a volume mesh with user prescribed boundary layer sizing normal to appropriate boundary faces. Typically, the anisotropic size field is local to the portion of the boundary it is prescribed on. This decoupling of the surface and volume meshing process, and lack of global anisotropic size information has several limitations. It fails to account for the impact of boundary layers in close proximity of each other. For instance, there may be imposed anisotropy on the surface mesh due to anisotropy of a boundary layer nearby. The decoupling of surface and volume meshing may result in lack of smoothness in the mesh when two boundary layers with different anisotropic directions are in close proximity (Fig. 1.1). It may also prematurely halt the smooth transition of boundary layer to an isotropic state when it encounters nearby geometry surfaces with no prescribed boundary layer. (Fig. 1.2). This leads to lack of sizing smoothness with abrupt change in mesh size [10].

Besides user prescribed boundary layer anisotropy, there may also be strong implicit anisotropy in the surface mesh induced by curvature. There is a strong coupling between the volume anisotropy of the boundary layer with surface curvature. The authors have analyzed this coupling for two-dimensions [4]. In three-dimensions, this may induce tangential refinement of the surface mesh due to the radiation of the small size field normal to the surface. Impact of this interaction on solution quality was studied in [25]. It is shown that gradient reconstruction using non-smooth anisotropic meshes works well in the absence of boundary curvature. It breaks down when curvature is present.

Inclusion of constraints like periodically matched meshes on portions of the boundary represent additional difficulties in boundary layer mesh generation process unless the surface mesh is generated first with proper accounting for the global impact of explicitly prescribed and implicit curvature-induced anisotropic sizes.

In a series of papers ([8,4,5,7]), the authors developed a framework for robust boundary layer mesh generation for complex geometries with completely general user prescribed sizing fields. It provides a solution for the systematic resolution of multiple boundary layer interactions by viewing the growing boundary layers as a discrete representation of the solution of the Eikonal equation [31] which models the evolution of normal offsets of boundary faces where boundary layers are prescribed. The approach at its core is an advancing-front scheme with the proper utilization of multiple normals at convex corners of the boundary, tangential adaptivity in the boundary layers, topological collapsing for colliding boundary layers due to concave geometric features and smooth sizing fields. The developed algorithms use smooth sizing fields constructed using *sources* [19,9]. *Sources* are associated with simple geometric entities such as points, lines, triangles and tetrahedra that are used to model the continuous variation of the sizing field throughout the domain. The ability to globally represent a smooth size field (distribution) is a key capability. It enables one to look at a mesh as a discrete class equivalence of a smooth sizing field for a given geometry [20] thus decoupling the representation of a sizing field from the specifics of a particular meshing algorithm. It provides the mesh generation and adaptation algorithms an explicit measure of how well they satisfy the required sizing field everywhere in the domain. This frees the user from having to understand low-level details specific to a meshing algorithm and instead specify sizing in an algorithm-agnostic manner and generate meshes that they expect without lots of trial and error. In the author's approach, users have several *a-priori* sizing options. Sizing attributes may be associated with specific topological entities in the model, based on curvature, or simply constant with simple canonical shapes. Each of these prescriptions create *sources*. Each source models the continuous variation (radiation) of the sizing value associated with it throughout space. The choice of model changes the rate at which the source spreads, but all produce smooth variations by construction.

Legacy methods to build sizing fields use background grids [27,19], uniform Cartesian meshes [28], or adapted Cartesian meshes [6,13]. More recently, [11,12] present an approach where a volume mesh is obtained from the surface mesh bounding the volume of interest. This volume mesh is then used as a background grid for the sizing, but the surface to be meshed has to be closed and the sizing components can not overlap. All these methods are limited to isotropic sizing distribution. A smooth anisotropic sizing field was described in [1] as part of an adaptive process. An approach to represent anisotropic fields is described in [30], but the anisotropic sources there are interpolated from isotropic ones and do not radiate to provide a smooth spatial distribution. Another approach solely based on anisotropic point sources is presented in [15]. There is a lack of general methods to prescribe and model smooth anisotropic sizing fields *a-priori* for mesh generation.

In a previous paper [4], some sources designed to generate the anisotropic sizing field corresponding to boundary layers on surfaces were presented. However, for volume boundary layer mesh generation [5], the sizing used was smooth, but isotropic. This paper fills a gap by extending it to represent *a-priori* fully anisotropic and smooth sizing fields for use in volume boundary layer mesh generation. It accounts for the combined effect of all sources of anisotropy including, prescribed surface and volume boundary layers, and implicit anisotropy due to curvature. The three-dimensional anisotropic size field is modeled as a metric-tensor [14] representing the mesh sizes in three mutually orthogonal directions.

The main contribution of this paper is a general approach for smooth representation and efficient evaluation of the metric-tensor at any point in space from the input anisotropic boundary layer sizes modeled as sources. The use of *a-priori* boundary layer mesh generation is primarily in computation of flow-fields modeled by Reynolds Average Navier-Stokes (RANS) equations [19]. For many flow regimes, the anisotropy is often very strong, due to very small first-layer thickness, and confined close to the boundary surfaces. Furthermore, due to curvature of geometries, the trace (imprint) of the anisotropic metric may produce extremely small mesh sizing on the surface. As the smallest mesh edge size on or near the surface gets close to the tolerance used in the underlying geometry modeling system,³ it leads to numerical difficulties in the geometry evaluations. This reduces robustness of the overall meshing process. To address these aspects, a scheme to modulate the strength of the anisotropy close to the boundary is provided. Users have the ability to balance accurate representation of the anisotropy with computational effort needed to generate a mesh.

The rest of the paper is organized as follows. Section 2 discusses details of analytical modeling of sizing for boundary layers. Section 3 describes the computation of anisotropic metric tensors as a general solution for representing boundary layer sizing at any point in the domain. Section 4 presents the different types of sources to effectively capture user prescribed volume and surface boundary layers. Section 5 presents an approach for modulation of boundary anisotropy for practical applications related to RANS type flow computations. Section 6 illustrates the proposed method through various numerical examples.

2. Boundary layer sizing representation

Boundary layers encapsulate a special type of anisotropic sizing field. Using the Eikonal equation as mathematical framework, one can see that its solution represents the evolution of a distance field providing normals and isodistance surfaces

³ First-layer thickness or the smallest mesh edge size on the surface is often on the order of $[10^{-6}, 10^{-10}]$ of the dimension of the domain. This is close to the tolerance used by most geometry systems.

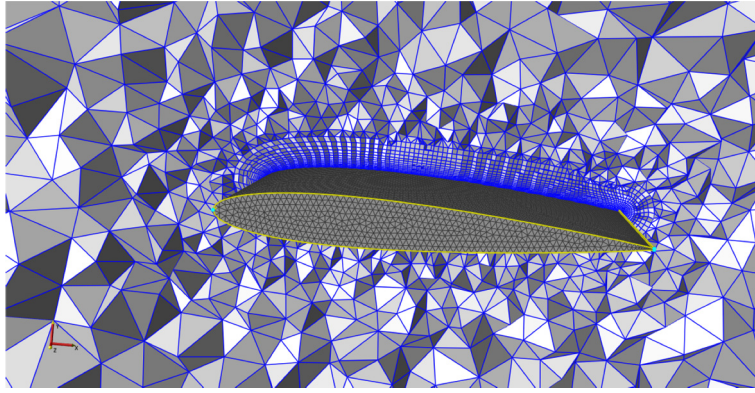


Fig. 2.1. The boundary layer normal sizing of the wing grows while the isotropic size also grows from the wing outwards. A jump is created in the mesh sizing since the sizing field itself is not smooth.

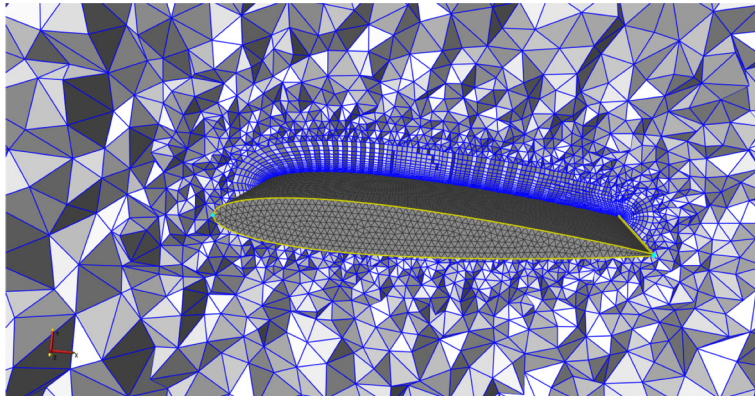


Fig. 2.2. The boundary layer normal sizing of the wing grows while the isotropic size is delayed from the wing outwards. No jump is created in the mesh sizing since the sizing itself is smooth.

(offsets) that decompose the sizing field in directions normal and tangential to the boundary. Boundary layer specification at any point only requires specifying the size field in the normal direction at the boundary that is closest to the given point. The size in the tangential direction is naturally induced based on the overall isotropic size field or the combined influence of other boundary layers or curvature induced sizes.

The typical specification of the boundary layer sizing involves user prescribed parameters such as the minimum (first) layer thickness and the growth rate for subsequent layers. Usually, it also includes an isotropic growth rate away from the surface. While this appears sufficient for a semi-structured approach for generating a boundary layer, it can often lead to lack of robustness and smoothness in the size field of the resulting mesh. To understand the distribution of mesh size normal to the surface, we look at different scenarios. In a first scenario, let us suppose that the user has set the same growth rate for the boundary layer and the isotropic field, depicted in Fig. 2.3(a). As a result, the first-layer thickness determines the smallest size even outside the boundary layer region because the anisotropic size remains smaller than the isotropic size everywhere. This is not optimal since the effect of anisotropy is expected to end outside the boundary layer region. Another issue appears for an isotropic sizing field. Here the isotropic prescription imposes a tangential size on the surface mesh which is extruded unmodified all the way up to the last layer of the boundary layer mesh. Since the isotropic tangential size increases away from the boundary layer surface, there is a jump in the tangential size at the transition from boundary layer into the isotropic region. A boundary layer mesh around a simple wing shows this problem in Fig. 2.1 where the boundary layer grows and can not match the isotropic sizing creating a jump in size at the boundary layer outer surface.

In order to remedy such issues, some fixes have been proposed. One is the idea of an increasing growth rate parameter of the boundary layer. This situation is displayed in Fig. 2.3(b). This is acceptable assuming that both the size distributions intersect at some point hopefully close to the region where the boundary layer becomes isotropic. Nonetheless, it is not robust and introduces additional complexity in the prescription of the boundary layer sizing and further constrains the meshing process. Another possible solution that the authors considered is shown in Fig. 2.3(c). Using the concept of a radius, or scaling-length [19], the isotropic sizing could be delayed in space through an input parameter that will maintain the isotropic distribution constant within the distance of that radius. If the value of the radius is chosen as the total thickness of the boundary layer, the sizing is controlled by the anisotropic sizing in the boundary layer, and the isotropic sizing outside the boundary layer. This avoids any additional constraint on the design of the boundary layer mesh. When

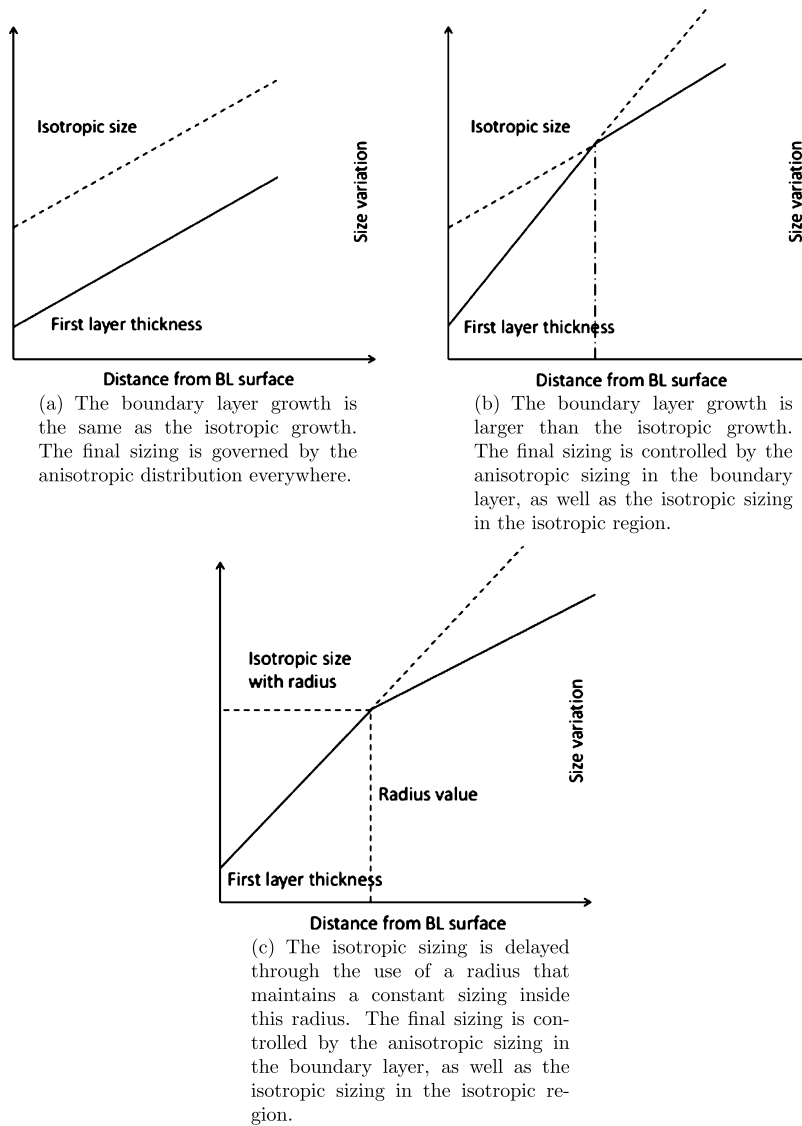


Fig. 2.3. Sizing distribution in the normal direction to the surface, for an isotropic size, and an anisotropic size (first layer thickness), with distinct growth rates.

used with an isotropic sizing field, it provides an improvement to the classic approach since it removes the size-jump in the tangential direction. Fig. 2.2 shows the resulting mesh for a wing.⁴

All the aforementioned ideas are *ad-hoc* in nature and work only when one knows the extent of the boundary layer region *a-priori*. This is not usually possible for complex geometries. Furthermore, all of the above approaches miss the fundamental nature of the anisotropic field due to curvature and its interaction with the anisotropy of the boundary layer sizing field itself. To address the robust representation of boundary layer anisotropy that ensures smoothness of the sizing field by design, one must use a general anisotropic sizing field to capture the boundary layer size. This is presented in the next section.

3. Anisotropic size computation

Anisotropy for mesh sizing is represented by a metric-tensor that captures the sizes in three mutually orthogonal directions [14]. The need to manipulate such tensor fields arises in many situations [33]. In the context of *a-posteriori* mesh adaptation, these metric-tensors are usually associated with the vertices of an existing mesh or a background mesh.

⁴ The original idea was not for use in boundary layer meshing. Its possible use for boundary layer meshing is first considered in this work.

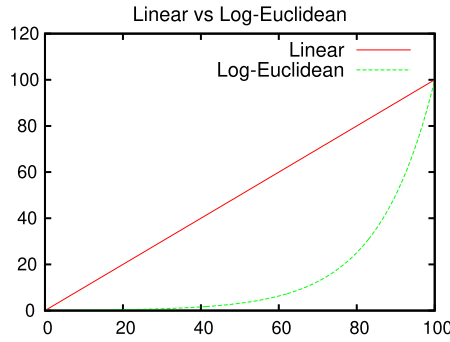


Fig. 3.1. Linear versus Log-Euclidean interpolation. When interpolating between two points, the log-Euclidean interpolation produces an interpolation which is very far from a linear function, which represents typical boundary layer prescription.

The metrics are then transferred to a new mesh using metric interpolation [21] and metric intersection [26]. Other approaches rely on metric decomposition using Cholesky factorization [16]. A closely related notion concerns the gradation of an anisotropic field [1,17].

There are several reasons one cannot use the existing approaches for use in *a-priori* generation of boundary layer meshes. First, the *a-priori* specification of boundary layer anisotropy is geometric, unlike those derived from *a-posteriori* error estimate. Preserving the eigenvector directions, in particular the one associated with the smallest size, is critical to the boundary layer mesh and cannot be modified or interpolated in a general way. Existing metric intersection procedures do not preserve the eigenvector directions [14]. A second issue arises from the use of log-Euclidean approximations [3] in the evaluating the metric field at a point. While this may be acceptable in an *a-posteriori* adaptation, it is not usable in representing anisotropy due to boundary layers where the normal size varies linearly with distance from the boundary. The log-Euclidean interpolation does not preserve the linear variation of the normal size away from the boundary needed in a boundary layer. This is illustrated in Fig. 3.1.

This paper presents a new approach to smoothly represent and evaluate anisotropic metric-tensors at any point in the domain. The approach explicitly captures and preserves the boundary layer size prescription. It consists of the following procedures:

1. Given a set of (an)isotropic sources, explicitly evaluate the anisotropic metric at any point in space. This is called the *sizing-proxy* algorithm.
2. Evaluate the anisotropic metric due to a specific source. This is called *source-modeling* algorithm.
3. Approximate the anisotropic metric over an axis-aligned bounding-box (AABB) to filter the sources that are used in the *sizing-proxy*. This is called the *min-max* algorithm.

The *sizing-proxy* is the interface by which any meshing algorithm would evaluate the smooth anisotropic size metric anywhere in space. *Source-modeling* is the actual mathematical description of how the metric field for a specific source evolves (radiates) in space. The *min-max* piece increases the computational efficiency by not having to deal with all the sources at every point in the *sizing-proxy* since usually only a few sources actually control the final metric at any point. Most of the sources have no significant influence on the final metric at a given point.

Algorithm 1 describes the *sizing-proxy*. It models the anisotropic metric associated with any source geometrically as an ellipsoid in three-dimensions according to a classic approach [14,1]. Given a point $\mathbf{X}$ belonging to the ellipsoid defined by a metric tensor $\mathbf{M}$, associated with a source, one can write

$$\mathbf{X}^T \mathbf{M} \mathbf{X} = 1. \quad (3.1)$$

The idea of the algorithm is to successively keep the eigenvectors associated with the smallest size in each dimension. In a first pass over all the sources, the smallest size and associated direction are kept. The orthonormal basis of the plane $\mathbf{P}$ normal to that direction is then built, defined by a pair of arbitrary, but mutually orthogonal vectors, $\mathbf{U}$ and $\mathbf{V}$. A vector from plane $\mathbf{P}$ then reads:

$$\mathbf{X} = a\mathbf{U} + b\mathbf{V}, \quad (3.2)$$

for some scalars coefficients a and b . Combining (3.2) and (3.1) gives:

$$a^2 \mathbf{U}^T \mathbf{M} \mathbf{U} + 2ab(\mathbf{U}^T \mathbf{M} \mathbf{V}) + b^2 \mathbf{V}^T \mathbf{M} \mathbf{V} = 1 \quad (3.3)$$

Diagonalizing the two-dimensional matrix associated with equation (3.3) yields the anisotropic sizes in the plane. A second loop over the sources will therefore provide the smallest size over $\mathbf{P}$. The last degree-of-freedom is then the direction normal to the two previous directions. The smallest size in that direction is finally computed using the metric for each source.

Algorithm 1 *Sizing-proxy*: get anisotropic metric at $p(x,y,z)$.**Require:** A series of isotropic and anisotropic sources.

```

1: Compute the smallest isotropic size  $s_{iso}$  of the isotropic sizes.
2: if There are no anisotropic sources then
3:   return  $s_{iso}$ 
4: end if
5: Find the smallest anisotropic size  $s_1$  and keep its eigenvector  $\mathbf{v}_1$ .
6: Build the plane normal to  $\mathbf{v}_1$ .
7: Compute the intersection of the normal plane with the ellipsoid associated with the metric for each source.
8: Find the smallest two-dimensional size  $s_2$ , along with its eigenvector  $\mathbf{v}_2$ .
9: Complete the three dimensional frame by computing the binormal  $\mathbf{v}_3$  to the two previous eigenvectors  $\mathbf{v}_1$  and  $\mathbf{v}_2$ .
10: Compute the smallest size along  $\mathbf{v}_3$ .
11: return  $\text{metric}(s_1, s_2, s_3, \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ 

```

The next section describes algorithms to do *source-modeling* for boundary layers and a *min-max* approximation of the minimum and maximum sizes over an AABB to efficiently determine those sources that are considered in the *sizing-proxy*.

4. Sources

The anisotropic sizing of the boundary layer is modeled using sources. For each source there are two key aspects. The first one models the value of the size at a given location in space. The second aspect provides an approximation of the source over an axis aligned bounding-box (AABB). Boundary layers prescribed on a geometry face and evolving into a volume will be called a volume boundary layer. Those prescribed on a geometry edge and evolving over a surface will be called a surface boundary layer. In this work the geometric basis for all boundary layer sources is simplices obtained from a tessellation of boundary faces. This means volume and surface boundary layers sources are associated with triangles and line segments on appropriate geometry faces and edges, respectively.⁵

A simple analytical model for a boundary layer at any point $\mathbf{x}$ is given by a linear representation of the form

$$s(\mathbf{x}, r, s_0) = (r - 1)d(\mathbf{x}) + s_0$$

where s is the size, $d(\mathbf{x})$ is the distance to the source, r is the growth ratio, and s_0 is the first layer height. More advanced forms that are high-order or use wall-laws [19] are also possible.

The sources, modeled as above, are all processed by an algorithm called the *sizing-proxy* whose job is to determine the final metric tensor using the sources that are relevant at a given location. The isotropic case is closely related to the lower envelope diagram [32]. For the anisotropic case, the final size computation is closely related to finding the smallest size in each direction, increasing drastically the complexity of the task at hand. Since a straightforward approach increases the complexity of the proxy algorithm significantly, an approximation of the source over an AABB is utilized.

A straightforward approach is to model and implement different surface and volume anisotropic sources independently. Apart from increasing drastically the total number of sources, it has the drawback that it may cause boundary layer induced anisotropy to bleed throughout the volume and not be limited to the boundary layer region. A better approach is to have a combined model of all the prescribed boundary layer anisotropies. This has the advantage of allowing for the clipping of the anisotropic metric as needed based on the isotropic size field thus producing a scalar size field in the domain beyond the boundary layer region. Such an approach retains the ability to have, if so desired, an anisotropic metric throughout the volume domain.

In the following subsections we present algorithms for boundary layer source models with increasing complexity:

1. Only volume boundary layer on a surface triangle (BLV-Source).
2. Volume boundary layer on surface triangle with one edge prescribed surface boundary layer (BLV-Source-Eikonal).
3. Volume boundary layer on surface triangle with multiple edges with prescribed surface boundary layer (BLV-Source-Eikonal-Multi).

In the first case, only the direction normal to the surface and the size in that direction is prescribed. The size in the plane tangent to the normal is induced by the isotropic size field. In the second case, one of the directions on the surface and its size is also prescribed. The third case is the most complex where two or more surface anisotropic fields are present and the effective directions and sizes along the surface must be computed. For each of these source models, two algorithms are presented. One evaluates the metric at a point in space and the other approximates the smallest and largest size produced by the source over an AABB for fast use in the *sizing-proxy*. This is very useful to compute the minimum of the maximum source sizes in the AABB (*min-max*) and discard any source whose minimum size is larger than the *minmax*. This prunes the list of potential sources that influence the size at a point greatly improving the speed at which the sizing field can be built.

⁵ One can use shapes other than triangles.

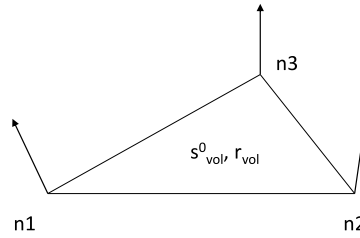


Fig. 4.1. A boundary layer volume source. A triangle is created with normal vectors at its vertices to represent the geometry. A first layer height s_{vol}^0 and a geometric ratio r_{vol} are associated with the source to represent a boundary layer sizing field.

4.1. Boundary layer volume source

The boundary layer volume source is the simplest type of source prescribed on a triangle depicted in Fig. 4.1. Its attributes consist of the normal vectors at the triangle vertices, first layer height, and the geometric ratio that controls its growth into the volume adjacent to the triangle. An algorithm to obtain the metric value at a point $\mathbf{p}(x, y, z)$ is given in Algorithm 2. It starts by finding the point on the source closest to $\mathbf{p}(x, y, z)$ and computing the normal, tangent and binormal vectors at the closest point. Then the distance of $\mathbf{p}(x, y, z)$ from the closest point is used to compute the radiated boundary layer and isotropic sizes r_{vol} and s_{iso} , respectively. The boundary layer size is clipped by the isotropic size. Accuracy of the computed metric depends on the computation of the normal vector. A hierarchy of approximations may be used to get vertex normal vectors. At the coarsest level, the normal vectors are approximated by the direction normal to the planar triangle. More accurate vertex normals are given by a quadratic triangle built from the edge midpoints on the geometry. When evaluating the size field very close to the geometry, it is best to use the normal from the actual mathematical description of the surface in the underlying geometry modeling system.

Algorithm 2 BLV-source::Get metric.

Require: A point $\mathbf{p}(x, y, z)$, a first layer height s_{vol}^0 , a geometric ratio r_{vol} , an isotropic size s_{iso}^0 , and an isotropic ratio r_{iso}

```

1: Get_Closest_Point( $\mathbf{p}$ , closest)
2: Compute normal  $\mathbf{n}$ , tangent  $\mathbf{tan}$  and binormal  $\mathbf{binorm}$ 
3: Get the radiated volume boundary layer size  $s_{vol}(s_{vol}^0, r_{vol})$  and the radiated isotropic sizes  $s_{iso}(s_{iso}^0, r_{iso})$  at  $\mathbf{p}$ 
4: if  $s_{vol} > s_{iso}$  then
5:   return  $s_{iso}$ 
6: else
7:   return  $\text{metric}(s_{vol}, s_{iso}, s_{iso}, \mathbf{n}, \mathbf{tan}, \mathbf{binorm})$ 
8: end if
 $\mathbf{binorm}$ 

```

Next, Algorithm 3, illustrates the computation of the smallest and largest metric over the AABB. Since the triangle may be curved, the normal direction carrying the smallest sizing varies. Therefore, the metrics at the three points of the triangle are intersected and unioned [26,14] to provide an approximation of the smallest and largest metrics at the source. Next the minimum and maximum metrics are determined for the AABB. This is done by approximating each AABB (cell) as a sphere and computing the distance from the source to the sphere. This distance is used to grow the minimum and maximum metrics at the source. The final values are determined by accounting for the isotropic size.

Algorithm 3 BLV-source::Get min-max metric on AABB.

Require: An AABB, its circumscribed sphere and its center $\mathbf{c}$

```

1: Get_Closest_Point( $\mathbf{c}$ , closest)
2: Get the min and max metrics at the source by taking into account the normals at the nodes and a first layer height  $s_{vol}^0$  in that direction
3: Grow the min and max metrics
4: Cut-off the eigenvalues by the isotropic values
5: return min and max metrics

```

4.2. Boundary layer volume source with one boundary layer edge

Building on the previous source, the effect of an anisotropic boundary layer growing on a surface from a boundary layer edge is taken into account. Due to the surface curvature, a boundary layer developing on the surface will not be based on the Euclidean distance, but rather on the geodesic distance to the emanating boundary layer edge. Some examples were presented in [4] where a dedicated source for the surface and for the volume was described. However, this unnecessarily duplicates the number of sources, and therefore the computational cost, to obtain the same resulting metric.

A source that combines a prescribed volume boundary layer with surface boundary on one edge, depicted in Fig. 4.2, is more efficient. A *fast marching method* (FMM) [31] is used on the surface to get the geodesic distance to the edge. Given the

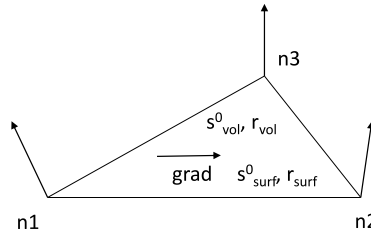


Fig. 4.2. A boundary layer volume source with boundary layer surface specification. A triangle is created with some normals at its vertices to represent the geometry. A first layer height s_{vol}^0 and a geometric ratio r_{vol} are associated with the source to represent a volume boundary layer sizing field, as well as a first layer height s_{surf}^0 and a geometric ratio r_{surf} to represent a surface boundary layer sizing field.

properties of the edge boundary layer, these distances are converted to sizes, and a three-dimensional surface gradient of the distance field is computed. The metric at each vertex is then computed with the surface sizing along the surface gradient direction, the first layer size in the normal direction, and the isotropic size specification along the binormal direction.

The algorithm to compute the anisotropic sizing for this source is given in Algorithm 4. Given a point in space, the algorithm first computes its closest point on the source and the normal vector at that point. The metric reference frame is completed by two other vectors. The second vector comes from the projection of the surface gradient onto the tangent plane. A third vector, the binormal, orthogonal to the first two vectors, completes the metric reference frame. On that frame, the volume boundary layer size is prescribed in the normal direction. The surface boundary layer size is prescribed along the projection of the surface gradient on the tangent plane. Finally, the radiated isotropic size is prescribed in the binormal direction.

Algorithm 4 BLV-Eikonal source::Get metric.

Require: A point $\mathbf{p}(x, y, z)$, a first layer height s_{vol}^0 , a geometric ratio r_{vol} , a linear surface sizing, an isotropic size s_{iso}^0 , and an isotropic ratio r_{iso}

- 1: Get_Closest_Point($\mathbf{p}, \text{closest}$)
- 2: Compute normal $\mathbf{n}$
- 3: Compute tangent $\mathbf{tan}$ from the gradient of the source, normal to $\mathbf{n}$
- 4: Compute $\mathbf{binorm}$ from $\mathbf{n}$ and $\mathbf{tan}$
- 5: Get the interpolated surface size at the closest point s_{surf}^0
- 6: Get the radiated surface size $s_{surf}(s_{surf}^0, r_{surf})$, the radiated volume anisotropic size $s_{vol}(s_{vol}^0, r_{vol})$, and the radiated isotropic size $s_{iso}(s_{iso}^0, r_{iso})$ at $\mathbf{p}$
- 7: **if** $s_{surf} > s_{iso}$ **and** $s_{vol} > s_{iso}$ **then**
- 8: **return** s_{iso} ;
- 9: **else**
- 10: **return** $\text{metric}(s_{vol}, s_{surf}, s_{iso}, \mathbf{n}, \mathbf{tan}, \mathbf{binorm})$
- 11: **end if**

For the *min-max* approximation, the same overall approach remains the same as described in Section 4.1. The metrics at each vertex of the triangle are computed, except that the boundary layer surface is taken into account in the creation of the metric at the vertices. These three metrics are intersected and unioned, giving rise to a minimum and maximum metric at the triangle. By approximating the AABB by a sphere, a minimum and a maximum distance are computed in order to grow these minimum and maximum metrics. The final algorithm is given in Algorithm 5.

Algorithm 5 BLV source::Get min-max metric on AABB.

Require: An AABB, its circumscribed sphere and its center $\mathbf{c}$

- 1: Get_Closest_Point($\mathbf{c}, \text{closest}$)
- 2: Get the min and max metric at the source by taking into account the normals at the nodes, a first layer height s_{vol}^0 in that direction, and the surface sizing at each node
- 3: Grow the min metric with the smallest ratio between the isotropic ratio and the volume ratio
- 4: Cut-off the eigenvalues by the isotropic values
- 5: **return** min and max metrics

4.3. Boundary layer volume source with various boundary layer edges

The most complex source represents multiple anisotropic boundary layer edges on the same surface, generating multiple anisotropic fields on the surface. That same surface may also have been selected to generate a volume boundary layer anisotropy. The geometry of the source is shown in Fig. 4.3. For each boundary layer edge belonging to such a source triangle, a FMM is used. The geodesic distances for each edge boundary layer are converted into sizes and a constant gradient vector is computed from each of these boundary layer edges.

In order to compute the metric at a given location, described in Algorithm 6, the closest point to the source and the normal vector is first obtained. Initial sizes are computed for each surface anisotropic field. The surface and volume sizes are

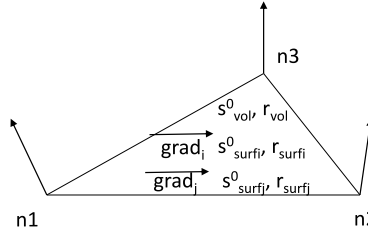


Fig. 4.3. A boundary layer volume source with multiple boundary layer surface specification. A triangle is created with some normals at its vertices to represent the geometry. A first layer height s_{vol}^0 and a geometric ratio r_{vol} are associated with the source to represent a volume boundary layer sizing field, as well as multiple first layer heights $s_{surf_i}^0$ and multiple geometric ratios r_{surf_i} to represent various surface boundary layer sizing fields.

radiated to the point of query. An orthonormal frame is computed using the normal vector, the gradient direction associated with the smallest surface sizing, and a binormal which is orthogonal to the other two. The size along the binormal is computed by knowing that the squared inverse of the minimum size is an eigenvalue of the metric and that the squared inverse of the second minimum size is associated with a given gradient. Finally, every size is clipped by the radiated isotropic size.

Algorithm 6 BLV-Eikonal-Multi source::Get metric.

Require: A point $\mathbf{p}(x, y, z)$, a first layer height s_{vol}^0 , a geometric ratio r_{ani} , an isotropic size s_{iso} , and an isotropic ratio r_{iso}

- 1: Get_Closest_Point($\mathbf{p}, \text{closest}$)
- 2: Compute normal $\mathbf{n}$, tangent $\mathbf{tan}$ and binormal $\mathbf{binorm}$
- 3: Get the interpolated sizes for each surface field, and sort the smallest two first, with their gradient and geometric ratios.
- 4: Get the radiated surfaces sizes s_{surf}^1, s_{surf}^2 , the radiated volumetric size s_{vol} , and the radiated isotropic size s_{iso} at $\mathbf{p}$
- 5: **if** $s_{surf}^1 > s_{iso}$ **and** $s_{surf}^2 > s_{iso}$ **and** $s_{vol} > s_{iso}$ **then**
- 6: **return** s_{iso} ;
- 7: **else**
- 8: Compute the size s_{surf}^3 on the binormal such that s_{surf}^1 is one size of the planar ellipse and s_{surf}^2 is the size of the planar ellipse in the direction of the second smallest gradient
- 9: **return** $\text{metric}(s_{vol}, s_{surf}^1, s_{surf}^3, \mathbf{n}, \mathbf{tan}, \mathbf{binorm})$
- 10: **end if**

The approximation of this source on an AABB is presented in Algorithm 7. The minimum and maximum metrics are computed by first computing the smallest metrics at the vertices of the triangle. For this source, the intersection and union process are repeated for each sizing field. These metrics are grown based on the smallest and largest geometric ratio associated with each surface and the volume boundary layer attributes.

Algorithm 7 BLV-Eikonal-Multisource::Get min-max metric on AABB.

Require: An AABB, its circumscribed sphere and its center $\mathbf{c}$

- 1: Get_Closest_Point($\mathbf{c}, \text{closest}$)
- 2: Get the min and max metric at the source by taking into account the normals at the nodes, a first layer height s_{vol}^0 in that direction, and every surface sizing at each node
- 3: Grow the min and metric
- 4: Cut-off the eigenvalues by the isotropic values
- 5: **return** min and max metrics

5. Modulation of the boundary layer anisotropy

In order to provide some flexibility, a user controlled input multiplicative parameter has been introduced that increases the first layer thickness strength in the sizing field, while maintaining a constant anisotropy until the sizing field reaches the desired anisotropic ratio. The sizing obtained is illustrated in Fig. 5.1, and is similar to the isotropic radius used in Fig. 2.3(c) in order to maintain a constant isotropic value. However, by adding a radius on the anisotropic value, a constant anisotropic ratio is maintained locally inside the given radius. The full strength of the boundary layer anisotropy is therefore not taken into account. However, this resolves potential issues on the surface due to geometric tolerances, and reduces the computational cost of the anisotropic sizing. Very close to the surface, the volume boundary layer generation will generate elements as expected. If the multiplicative factor is very large, the anisotropic size becomes larger than the isotropic size, and an isotropic field is generated, corresponding to a more traditional method. We call this multiplicative factor *boundary layer diffusion* parameter, since it diffuses the anisotropy.

In order to implement the anisotropic radius, the only modification consists of replacing the distance computation

$$s(x, r, s_0) = (r - 1)d(x) + s_0 \quad (5.1)$$

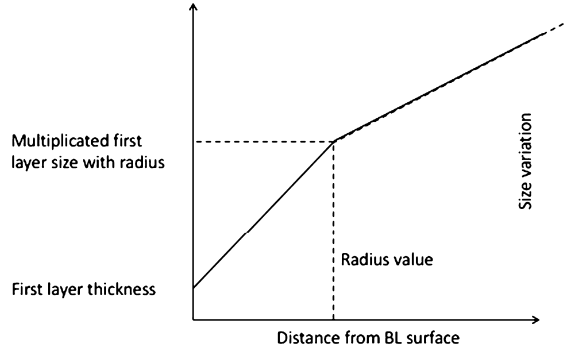


Fig. 5.1. Use of a multiplicative factor on the first layer thickness as well as an anisotropic radius.

by:

$$s(x, r, s_0, radius) = (r - 1) \max(0, d(x) - radius) + s_0 \quad (5.2)$$

Given the multiplicative coefficient μ , a radius is computed as

$$radius = s_0(\mu - 1)/(r - 1). \quad (5.3)$$

The anisotropic value of the sizing is constant up to the corresponding size in the boundary layer.

6. Examples

In this section, some examples are presented to illustrate the effect of the anisotropic sources on surface and volume mesh generation. The mesher receives a tensor at a given location in space. The unstructured surface and volume mesher use the eigenvector aligned point placement proposed in [23,22]. It therefore clearly illustrates the anisotropy of the sizing field. Furthermore, there is no isotropic or anisotropic curvature based sizing prescribed in the following examples. The curvature refinement appears naturally from the coupling between the volume boundary layer anisotropy and the natural surface curvature.

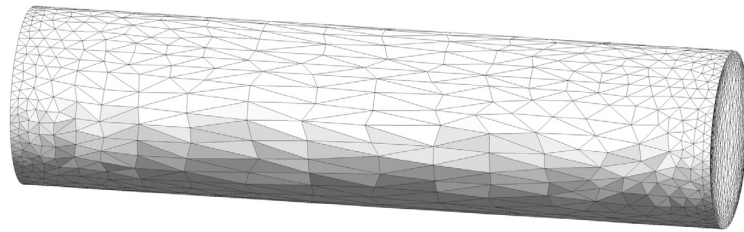
6.1. Cylinder

A cylinder is chosen as a simple geometry that presents inherent anisotropy in surface curvature. Since a sphere has constant curvature, as well as the same radii of curvature, the anisotropic field generated from a volume boundary layer on the sphere would degenerate to an isotropic one. In the first case, the cylindrical face is chosen to generate a boundary layer. The top edges of the cylindrical face are also selected for isotropic sources. Fig. 6.1 displays the surface mesh, while Fig. 6.2 display a cut of the volume mesh. The transition from isotropic to anisotropic is clearly seen in the axial direction. Also, the size in the radial direction in the middle of the cylinder is solely dictated by the boundary layer normal sizing coupled with the constant curvature of the cylinder.

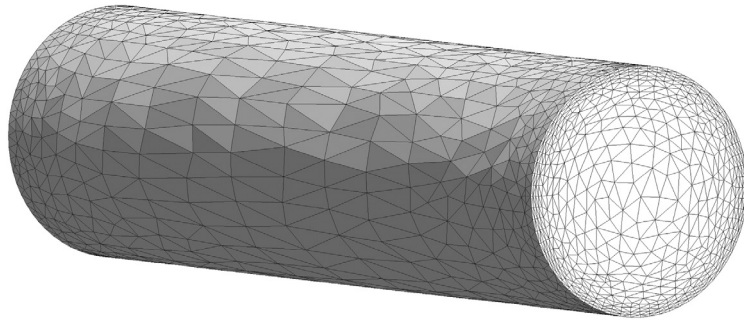
In the second case, the three faces are selected for volume boundary layer. This is the only user prescription in terms of sizing. The example is displayed in Fig. 6.3. The resulting anisotropic field is created by both end caps of the cylinder as well as the cylindrical face. The mesh is anisotropic with the smallest size in the axial direction close to the caps, and smoothly evolves to an anisotropy where the axial direction is the largest size close to the middle of the cylinder. Regarding the volume, the boundary layer is clearly seen inside the cylinder while the transition between the semistructured boundary layer and the anisotropic unstructured mesh is very smooth. It is worth noticing the complexity of the anisotropic sizing field with respect to the simplicity of input given by the three faces.

6.2. Multiplicative factor on the ONERA M6 wing

In order to illustrate the effect of the *boundary layer diffusion* factor proposed in section 5, the geometry of the ONERA M6 wing is used. A surface boundary layer is set on the trailing edges and a volume boundary layer is set on the wing surfaces. There is no other source specified. In particular, no sizing has been attributed to the surface. The boundary layer diffusion parameter is set from 1 to 1000. The effect of that parameter is shown in Fig. 6.4. For a value of 1000, some anisotropy is felt on the leading edge, due to the interaction between the volume boundary layer and the curvature of the surface, but the mesh is overall very coarse. When the value of the parameter is decreased, the anisotropy is more strongly felt at the leading edge.



(a) Surface mesh from the side of the cylinder



(b) Surface mesh from the top of the cylinder

Fig. 6.1. A cylinder with the cylindrical face selected for boundary layer. The edges are added as sources also. The transition from isotropic to anisotropic is clearly seen in the axial direction. Also, the size in the radial direction in the middle of the cylinder is solely dictated by the boundary layer.

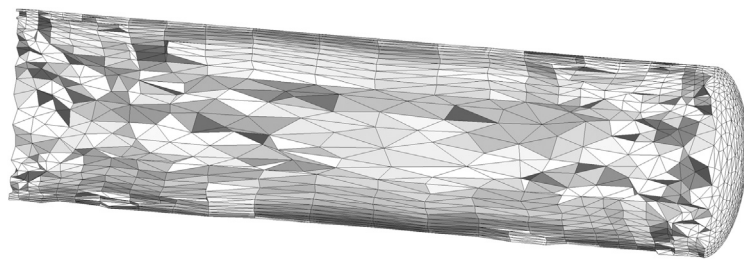
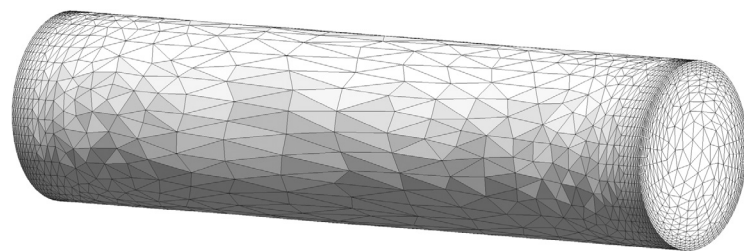
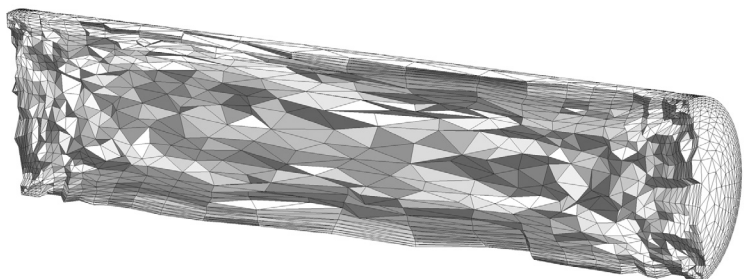


Fig. 6.2. Cut of the cylinder in the volume with the cylindrical face chosen for the boundary layer.



(a) Surface mesh



(b) Volume mesh

Fig. 6.3. A cylinder with every surface selected for boundary layer.

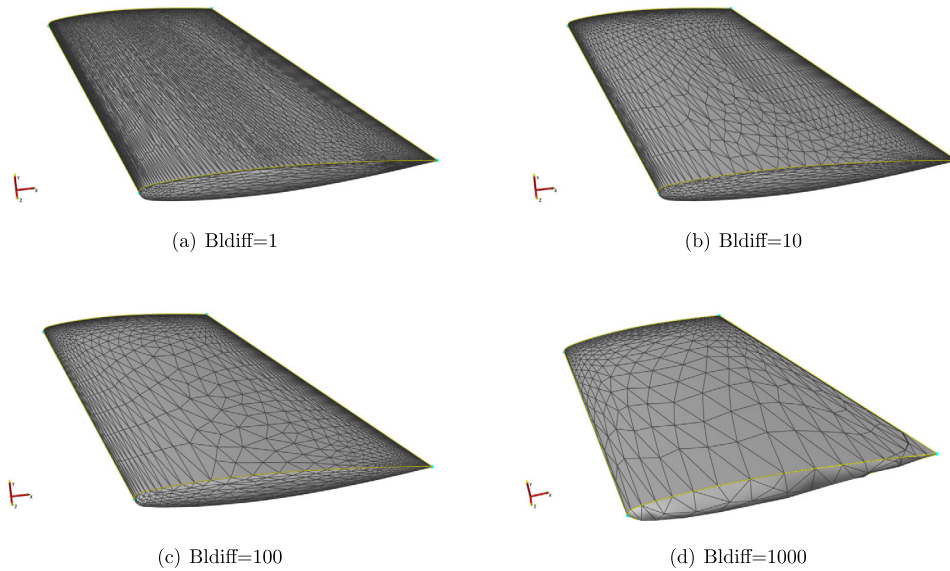


Fig. 6.4. The ONERA M6 wing with different values for the boundary layer diffusion.

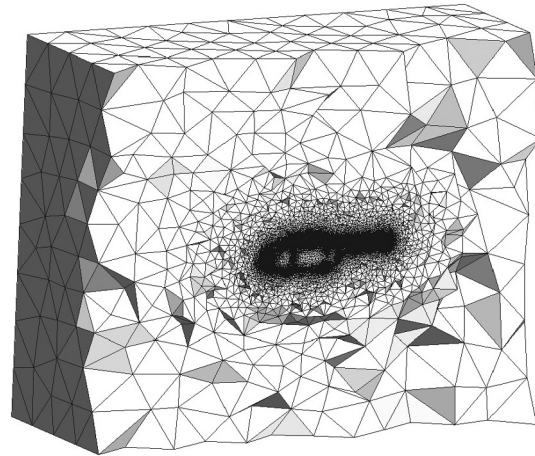


Fig. 6.5. Global view of the Robin model.

6.3. Robin body

The Robin body is presented as a more realistic geometry. The meshes for this are displayed in Figs. 6.5 and 6.6. Some isotropic sources were set around the geometry edges of the model to produce a more complicated sizing field. A boundary layer edge was set at the concave location near the rotor. From a user standpoint, sizing specification requires a minimal effort on a workstation to realize smooth anisotropic sizing automatically. The surface mesh shows the interaction between the boundary layer and curvature anisotropy as the mesh aligns along the principal directions of curvature of the surface. A smooth transition between the anisotropic regions to the more isotropic areas closer to the edges is also seen. Finally, the transition between the semistructured and the unstructured mesh is seamless since a smooth gradation is built by construction. Fig. 6.7 shows that the new approach enables smooth propagation of the anisotropic sizing along the surface whereas a legacy semi-structured approach results in the abrupt termination of the boundary layer when the neighboring face has no anisotropic size information. This creates a size jump depicted in Fig. 1.2. The globally smooth anisotropic sizing field enforces the correct anisotropy independent of the meshing approach.

6.4. Multi-element airfoil

This example illustrated in Fig. 6.8 shows a classic situation exhibiting the flaps and slats of an airfoil. The multiple elements are close enough to mutually influence the boundary layer they generate and interact with the boundary layer from the wing. Traditionally, the surface mesh is generated with an isotropic sizing. However, in a case like this, the surface

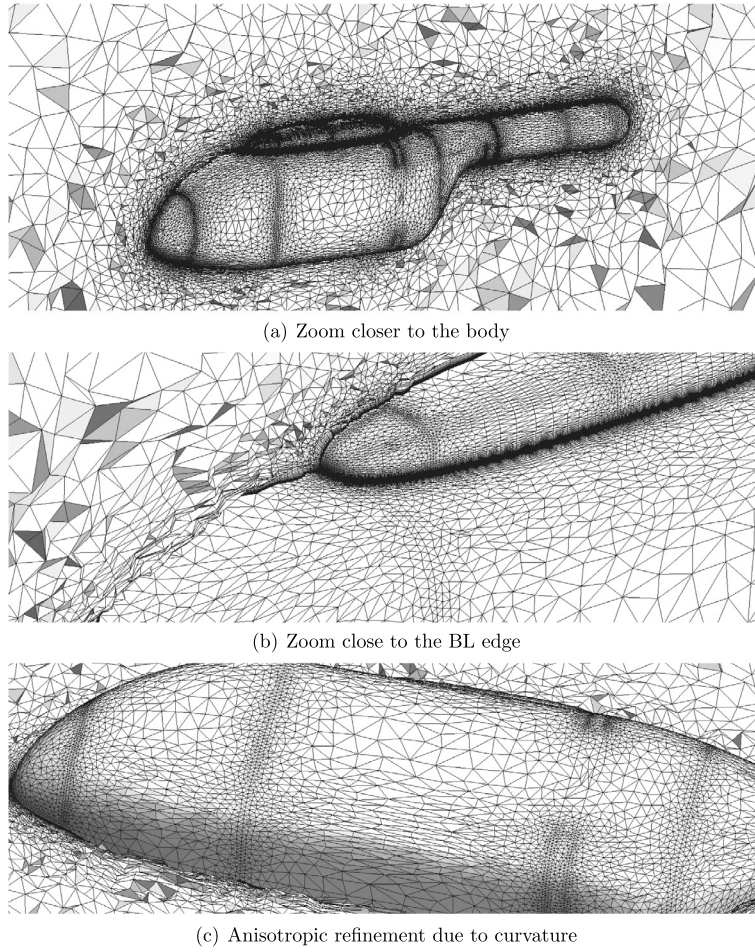


Fig. 6.6. The Robin body.

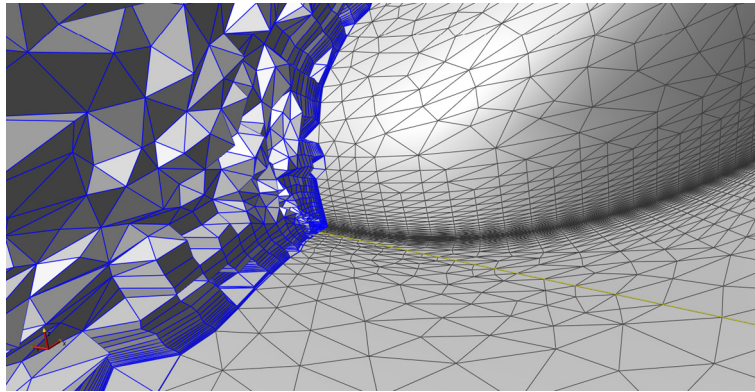


Fig. 6.7. The anisotropic field radiates everywhere in space. The semistructured mesh has been stopped by a geometric edge, shown in yellow in the picture. However, the mesh is still anisotropic beyond the geometric edge thanks to the anisotropic field. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

mesh itself is influenced by the volume boundary layer. If the surface mesh is generated independently from the influence of the boundary layer, it is not possible to create a smooth transition since the surface mesh has already been generated, and the proximity is too strong to generate a smooth transition.

From a user standpoint, the only specifications imposed are the isotropic size on the surface as well as the boundary layer first thickness and growth coefficient. Based only on that input, a complex and smooth anisotropic sizing field is generated.

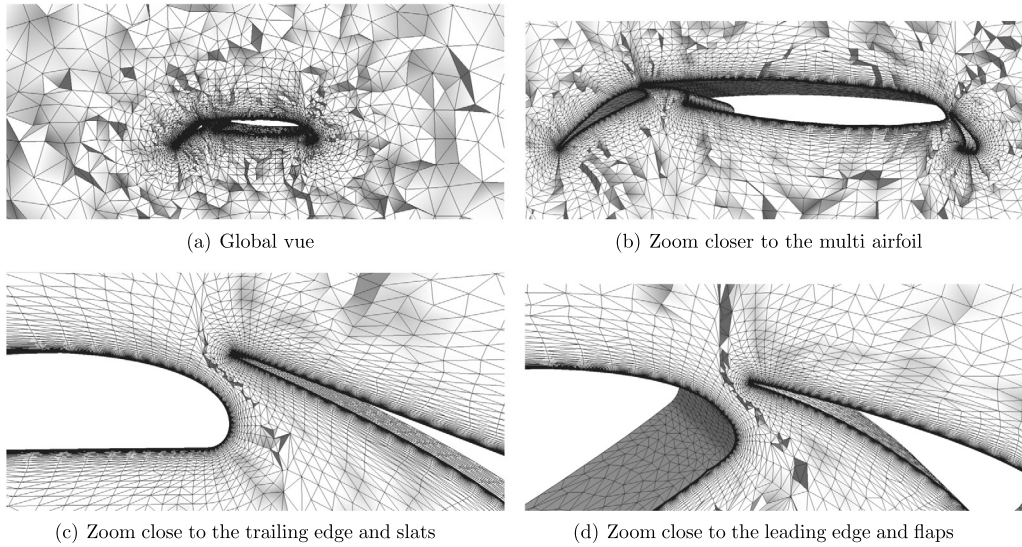


Fig. 6.8. The HL-CRM 2d extruded body.

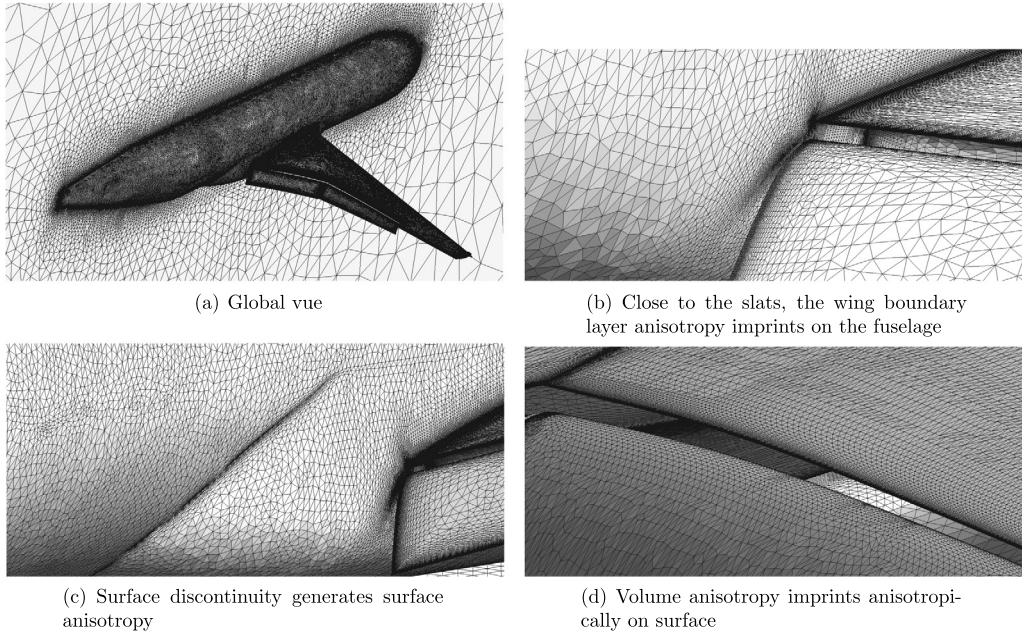


Fig. 6.9. The HL-CRM model.

6.5. High lift surface mesh

The more realistic surface of the High Lift geometry is used here and illustrated in Fig. 6.9. The model is relatively complex, and numerous small gaps between the different parts occur, that may stop the boundary layer growth and expose fronts with very different length scales. Thanks to the proposed method, the surface mesh is already compliant with the boundary layer surface and volume anisotropy. Even if the semistructured method stops because of proximity, the exposed fronts will present a smoothness inherited from the global anisotropic sizing field smoothness. For example, it is remarkable that the wing boundary layer imprints on the fuselage, which is never reached by generating an isotropic sizing field, as illustrated in Fig. 6.9. It is also clearly seen how the discontinuity on the fuselage coupled with the volume anisotropy creates surface anisotropy. Finally, the anisotropy observed on the front of the flap coming from wing boundary layer validates the anisotropic sizing field, since mesh alignment is inherited from the metric eigenvectors. This allows any volume mesher algorithm to generate smooth elements as long as the sizing is fulfilled.

7. Conclusion

The main contribution of this work consists of an approach to represent and efficiently evaluate the smooth anisotropic sizing fields due to *a-priori* specification of boundary layers on surfaces and in volumes. This is achieved using sources that model the anisotropic sizing metric induced by interaction of the surface and volume boundary layer specification. Unlike legacy approaches, a globally smooth sizing field is constructed before the mesh process begins making both the specification and evaluation of the anisotropic metric independent of the specific approach employed in generating a boundary layer. By design, this couples the anisotropic metrics from surface and volume boundary layers with the curvature induced anisotropy.

Three types of sources were described. One for a volume boundary layer, a volume boundary layer with a surface boundary layer, and a volume boundary layer with multiple surface boundary layers. These sources have been implemented efficiently to model any general boundary layer specification. The introduction of a *boundary layer diffusion* parameter (radius) in sources has multiple benefits. In an isotropic sizing context, it minimizes the possibilities of a jump in sizes the tangent directions. In an anisotropic sizing context, it allows the anisotropic sizing to be restricted to the boundary layer region, preventing anisotropic radiation throughout the domain. This also enables stable evaluation of the boundary geometry by preventing the smallest mesh edge sizes to fall below the geometric tolerance of the underlying geometry system.

Numerical examples illustrated the accuracy and novelty of the proposed method, which guarantees a smooth mesh *in every direction* and a seamless transition between the semi-structured and unstructured part as a natural result of the method, while reducing to a bare minimum the user input to achieve it.

CRedit authorship contribution statement

Romain Aubry Conceptualization, Methodology, Software Saikat Dey Writing Others Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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